```
  public UserController(Database database) {
    this.database = database;
  }

  public JSONObject registerUser(Request request,
      Response response) throws Exception {
    var json = new JSONObject(request.body());
    var username = json.getString("username");
    var password = json.getString("password");

    if (!username.matches(USERNAME_PATTERN)) {    #A
      throw new IllegalArgumentException("invalid username");
    }
    if (password.length() < 8) {
      throw new IllegalArgumentException(
          "password must be at least 8 characters");
    }

    var hash = SCryptUtil.scrypt(password, 32768, 8, 1);
#B
    database.updateUnique(    #C
        "INSERT INTO users(user_id, pw_hash)" +
        " VALUES(?, ?)", username, hash);

    response.status(201);
    response.header("Location", "/users/" + username);
    return new JSONObject().put("username", username);
  }
  ..
}
```

#A Apply the same username validation that you used before.
#B Use the Scrypt library to hash the password. Use the recommended parameters for 2019.
#C Use a prepared statement to insert the username and hash.